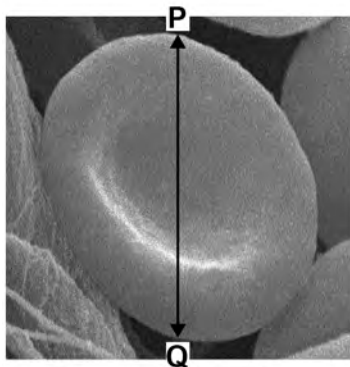


0 2 . 1

Figure 2 shows an image of a red blood cell at a magnification of $\times 5500$

Do not write
outside the
box

Figure 2



Calculate the actual diameter in μm of the red blood cell between points **P** and **Q**.

Show your working.

[2 marks]

Answer _____ μm



0	2	.	2
---	---	---	---

A haemocytometer is a special microscope slide that can be used to determine the mean number of red blood cells in 0.004 mm^3 of blood.

- A researcher prepared a 10^{-3} dilution of a sample of blood from an adult.
- Using a haemocytometer, the researcher determined that the mean number of red blood cells in 0.004 mm^3 of the diluted blood sample was 21

The volume of blood in the body of the adult was 4.8 dm^3

Calculate the total number of red blood cells in the body of this adult.

Show your working.

[2 marks]

Answer _____

0	2	.	3
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The solution used to dilute the blood had to have the same water potential as the blood.

Explain why.

[2 marks]

Question 2 continues on the next page

Turn over ►



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02.4

There are four main blood groups in the human ABO blood group system.

Table 1 shows the basis on which each of these blood groups is classified.

Table 1

ABO blood group	Antigens present on red blood cells	Antibodies present in blood plasma
A	A	Anti-B
B	B	Anti-A
AB	A and B	No anti-A and no anti-B
O	No A and no B	Anti-A and anti-B

A transfusion of blood from a blood group **A** donor to a blood group **B** recipient would cause agglutination of the donated red blood cells.

Use information in **Table 1** to explain why.

[2 marks]

02.5

Using **Table 1**, give the blood groups of people who could accept a donation of blood group **O** without causing agglutination of the **donated** red blood cells.

[1 mark]

Blood groups

9

